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# Can OCR Rules Replace Azure OpenAI?

Medical receipt extraction benchmark

9 receipts · 6 fields · rules vs AI extraction

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# Executive Summary

OCR + rules is not enough to replace AOAI across varied receipts.

**96.1%**

**Azure DI + AOAI (prompt v2)**

*Development result*

**51.0–70.6%**

**OCR + multilingual rules**

51.0% PaddleOCR Light · 70.6% Azure DI

*Retrospective baseline*

## Why AOAI may still be needed

- Can often handle unfamiliar layouts without new rules (not guaranteed)
- Can potentially extract more fields for analytics (not yet benchmarked)
- Rules remain fine for simple, clearly labelled checks


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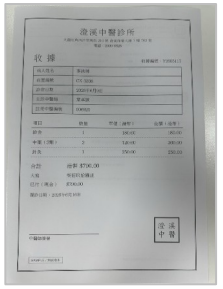
# What We Compared

For each of the 9 receipts, we extract 6 key fields and compare the results across 3 approaches.

| | 1. Read the text (OCR) | | 2. Find the fields | | |
|---|---|---|---|---|---|
| **PaddleOCR Light + rules** | **PaddleOCR Light**<br>open source, own compute | → | **Hand-written rules** | → | **6 fields** |
| **Azure DI + rules** | **Azure Document Intelligence**<br>cloud OCR | → | **Hand-written rules** | → | **6 fields** |
| **Azure DI + AOAI** | **Azure Document Intelligence**<br>cloud OCR | → | **Azure OpenAI instructions** | → | **6 fields** |


English


Bilingual


Chinese

**9 receipts**: 7 printed test receipts (English, bilingual, Chinese layouts) and 2 from the first benchmark round.

**6 fields**: doctor, registration no., patient, receipt no., service date, total.

*51 of 54 possible fields were scored. 3 receipts do not contain a doctor registration number, so those fields were excluded.*

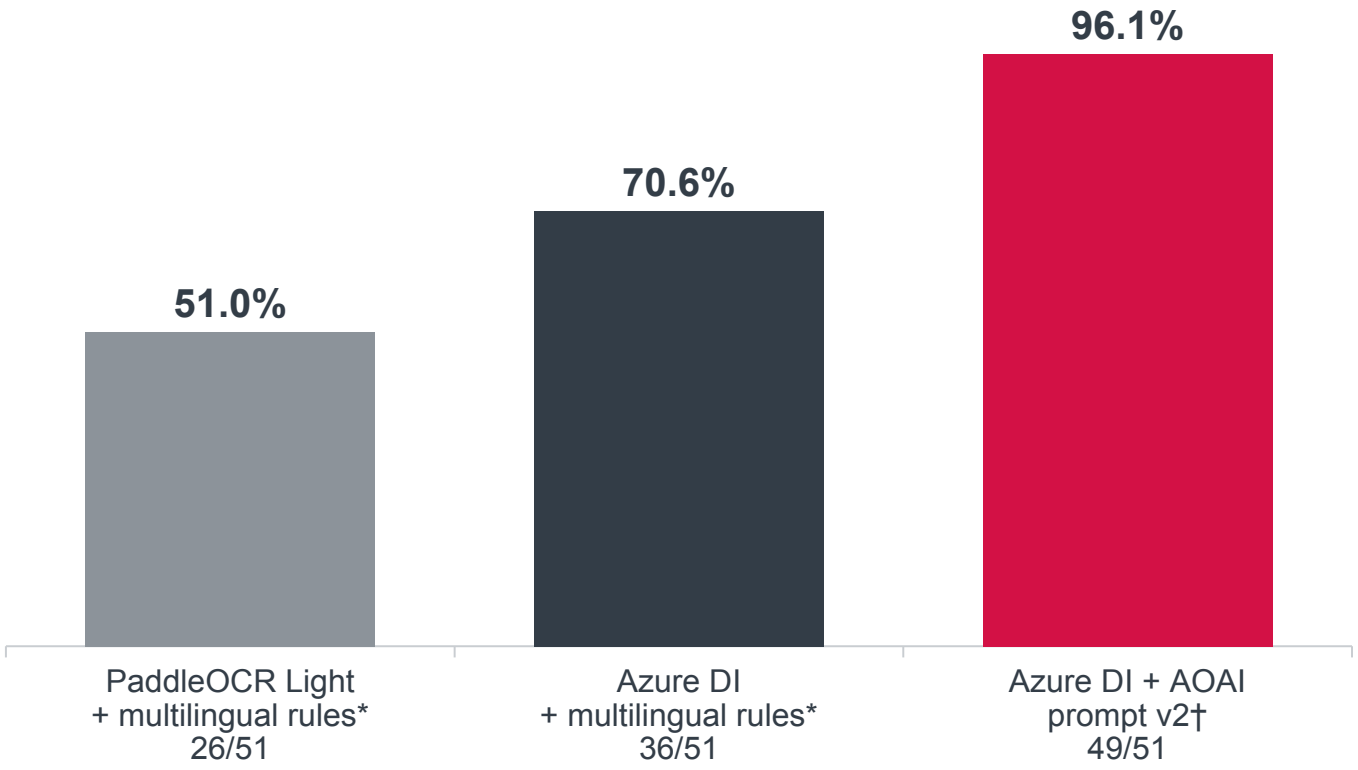

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# Rules vs AOAI on the Same Receipts

MEASURED · DEVELOPMENT SET

Multilingual OCR rules reached 51.0–70.6%; Azure DI + AOAI reached 96.1%.




**Multilingual rules***

- General rules with common Chinese labels
- No receipt-specific rules

**What explains the gap**

- PaddleOCR misread labels (slide 5)
- Rules missed context that AOAI used (slide 6)

** Retrospective baseline  † Development result (prompt refined on these receipts)  Neither is independent validation. 9 receipts, 51 fields.*

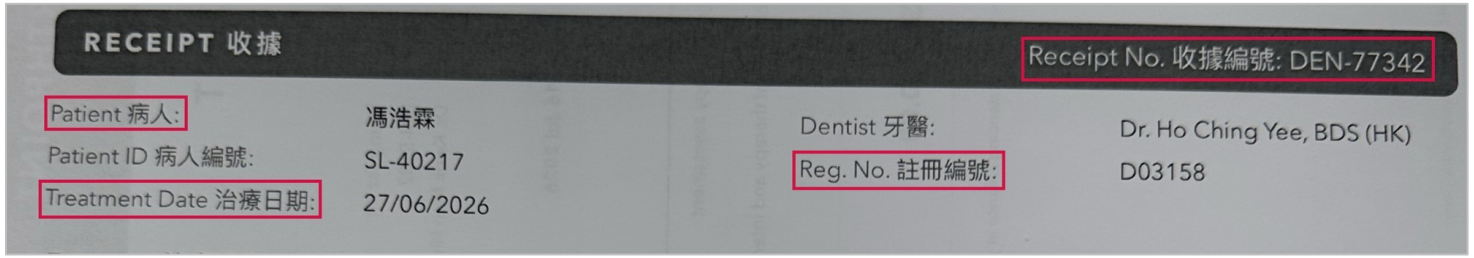

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# Why Azure DI Beat PaddleOCR Light

MEASURED · RETROSPECTIVE

Same rules on both. PaddleOCR misread the labels, so the rules could not find the values.


RECEIPT 收據

Receipt No. 收據編號: DEN-77342

Patient 病人: 馮浩霖

Patient ID 病人編號: SL-40217

Treatment Date 治療日期: 27/06/2026

Dentist 牙醫: Dr. Ho Ching Yee, BDS (HK)

Reg. No. 註冊編號: D03158

**10 extra errors with PaddleOCR**

All from reading the label text: words run together, wrong characters, a misread HK$ sign. None from the rules or the layout.

On this receipt PaddleOCR lost 4 fields: patient, registration no., treatment date and receipt no.

| Field | Correct value | PaddleOCR read the label as… | Azure DI read it as… |
|---|---|---|---|
| **Registration no.** | D03158 | ✗ Reg.No. 註冊編號 *words run together → not found* | ✓ Reg. No. 註冊編號 → *found* |
| **Treatment date** | 27/06/2026 | ✗ TreatmentDate 治療日期 *words run together → not found* | ✓ Treatment Date 治療日期 → *found* |
| **Receipt no.** | DEN-77342 | ✗ Receipt No. 收據编號 *编 read for 編 → not found* | ✓ Receipt No. 收據編號 → *found* |

*Same multilingual rules (retrospective baseline) on both engines. Receipt 5 is a bilingual dental receipt.*

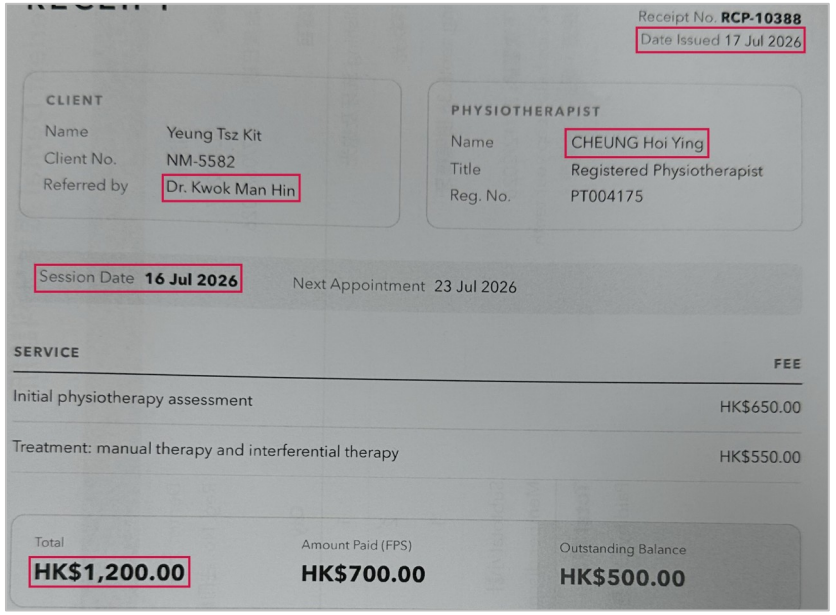

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# Why AOAI Beat the Rules

**MEASURED · RETROSPECTIVE**

Both read the same Azure DI text. AOAI used the context; the rules followed fixed patterns.




| Field | Correct | Rules | AOAI |
|---|---|---|---|
| **Practitioner**<br>Receipt 6 | CHEUNG Hoi Ying | ✗ Dr. Kwok Man Hin<br>*took the referring doctor* | ✓ CHEUNG Hoi Ying |
| **Service date**<br>Receipt 6 | 16 Jul 2026 | ✗ 17 Jul 2026<br>*took the date the receipt was issued* | ✓ 16 Jul 2026 |
| **Total**<br>Receipt 6 | HK$1,200 | ✗ not found<br>*could not read the card layout* | ✓ HK$1,200 |
| **Total**<br>Receipt 5 | HK$1,575 | ✗ HK$1,750<br>*added up the items, ignored the discount* | ✓ HK$1,575 |

**Rules: logic written case by case**

Unfamiliar labels, layouts or scenarios may need additional rules and retesting (tuning these 9 grew the rules from 69 to 225 label phrases).

**AOAI: plain-language instructions**

Can often handle varied wording without new code (not guaranteed); one prompt revision fixed 4 fields and broke none.

*AOAI was right with its original prompt (written before these receipts) and with prompt v2. Rules = Azure DI + multilingual rules (retrospective baseline).*


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# The Doctor Database Approach

PROPOSAL · NOT BENCHMARKED

Exact matching can identify registered doctor names in OCR text, but cannot determine who treated the patient.

## The proposal

1. **OCR reads the receipt**
   Azure DI or PaddleOCR
2. **Match against the registry**
   ~40,000 doctors · ~80,000 English and Chinese name variants
3. **Strict exact match**
   No fuzzy matching, no AI

**Straightforward; no experiment needed**
Low cost · Deterministic · Easy to verify

**Only works if** the OCR reads the name exactly and the registry contains that practitioner.

## But a name match is not the treating doctor

| Seen in our receipts | |
|---|---|
| **Referring doctor** | Receipt 6: “Referred by Dr. Kwok Man Hin”; the treatment was by a physiotherapist (also receipt 2) |
| **Several doctors or staff** | Receipt 8: a doctor and a nurse. Vet invoice: a clinician and a signing vet |
| **OCR misreads the name** | Signature “Dr. Lok Yi Choi” read as “Dr. Lok Vichol”: no exact match |

| Hypothetical (not in our receipts) | |
|---|---|
| **Patient is a registered doctor** | *The patient's own name matches the registry* |
| **Clinic director in the header** | *A director's name printed in the letterhead* |

*AOAI reads context such as “Referred by” (slide 6) but has not been tested on these cases. Names on our test receipts are fictional, so registry match rates were not measured.*


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# Advantages and Disadvantages

Rules win on cost, auditability and predictability; AOAI on accuracy, maintenance and flexibility.

| | OCR + rules | Azure DI + AOAI | Evidence |
|---|---|---|---|
| **Cost** | + PaddleOCR: own compute, no API fee (Azure DI: page fee) | − Azure DI page fee + ~962 input / 243 output tokens per receipt | *Prices to confirm* |
| **Speed** | PaddleOCR 13.9 s (CPU) · Azure DI 3.8 s | 6.4 s per receipt | Measured |
| **Accuracy** | − 51.0–70.6% with multilingual rules (retrospective) | + 96.1% with prompt v2 (development result) | Measured |
| **Maintenance** | − Unfamiliar labels or layouts may require additional rules and retesting | + Many changes made in plain-language instructions; still retest | *Measured / expected* |
| **Flexibility** | − Finds only what it was written for | + New fields can be requested in the prompt (not yet benchmarked) | *Expected* |
| **Auditability** | + Every result traceable to a rule | − Harder to explain; keep quoted evidence and logs | *Expected* |
| **Reliability** | + Same input, same output | − Can return a confident wrong value (e.g. a sub total as the total); model updates need retesting | *Measured / expected* |

*Green = stronger side. Multilingual rules include common Chinese labels (retrospective baseline). Expected = not measured here.*


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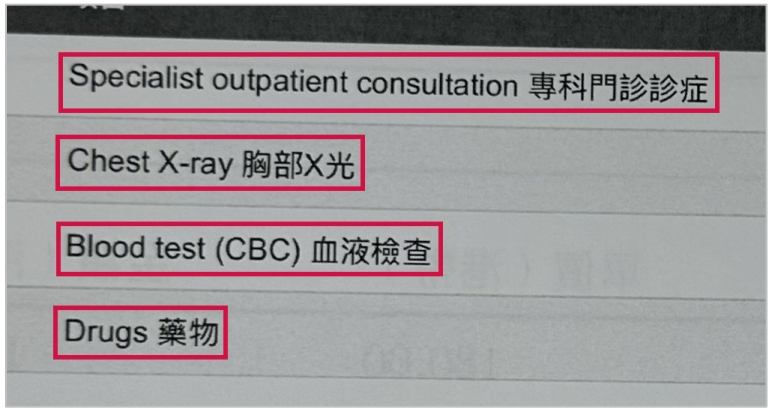
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# Beyond Six Fields: From Receipts to Insights

POTENTIAL · NOT YET TESTED

Today we extract 6 fields. AOAI could also capture what was treated, tested and prescribed.

### 1. What the receipt shows




### 2. What AOAI could extract

| Item | Type | HK$ |
|---|---|---|
| Specialist consultation | **Consultation** | 950.00 |
| Chest X-ray | **Test** | 620.00 |
| Blood test (CBC) | **Test** | 380.00 |
| Drugs | **Medication** | 245.50 |

*Illustration typed from the receipt text, not AOAI output*

### 3. What we could analyse

**Cost by treatment**
Spend on consultations, tests and medicines

**Provider comparison**
Typical charges per test across providers

**Trends over time**
Which treatments and medicines are rising

**Unusual claims**
Combinations or amounts that stand out

**Potential, not yet tested.** Only the 6 current fields have been benchmarked. Diagnoses, symptoms, treatments and medications would need a pilot.


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# Recommendation

Use rules for simple, labelled checks. Use AOAI where receipts vary or richer data is needed.

### Simple OCR rules are sufficient when…

- Layouts are few, known and stable
- Fields have clear printed labels
- The question is simple, e.g. is a diagnosis comment present?
- No API fee and fully traceable output are required

### AOAI adds real value when…

- Providers, layouts and languages vary (Chinese, bilingual)
- Fields lack consistent labels: names with titles, totals vs subtotals
- We want less scenario-specific code to maintain
- Richer medical data is wanted (potential, not yet benchmarked)

### 1. Freeze

the rules and AOAI prompt v2

### 2. Validate

30+ unseen, real, anonymised receipts; count rule changes

### 3. Pilot extra fields

medications, treatments, symptoms, diagnoses with ground truth

### 4. Confirm cost

Document Intelligence pages and gpt-5-mini tokens

*Evidence so far: 9 receipts; results are development or retrospective, not yet validated on unseen receipts.*


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